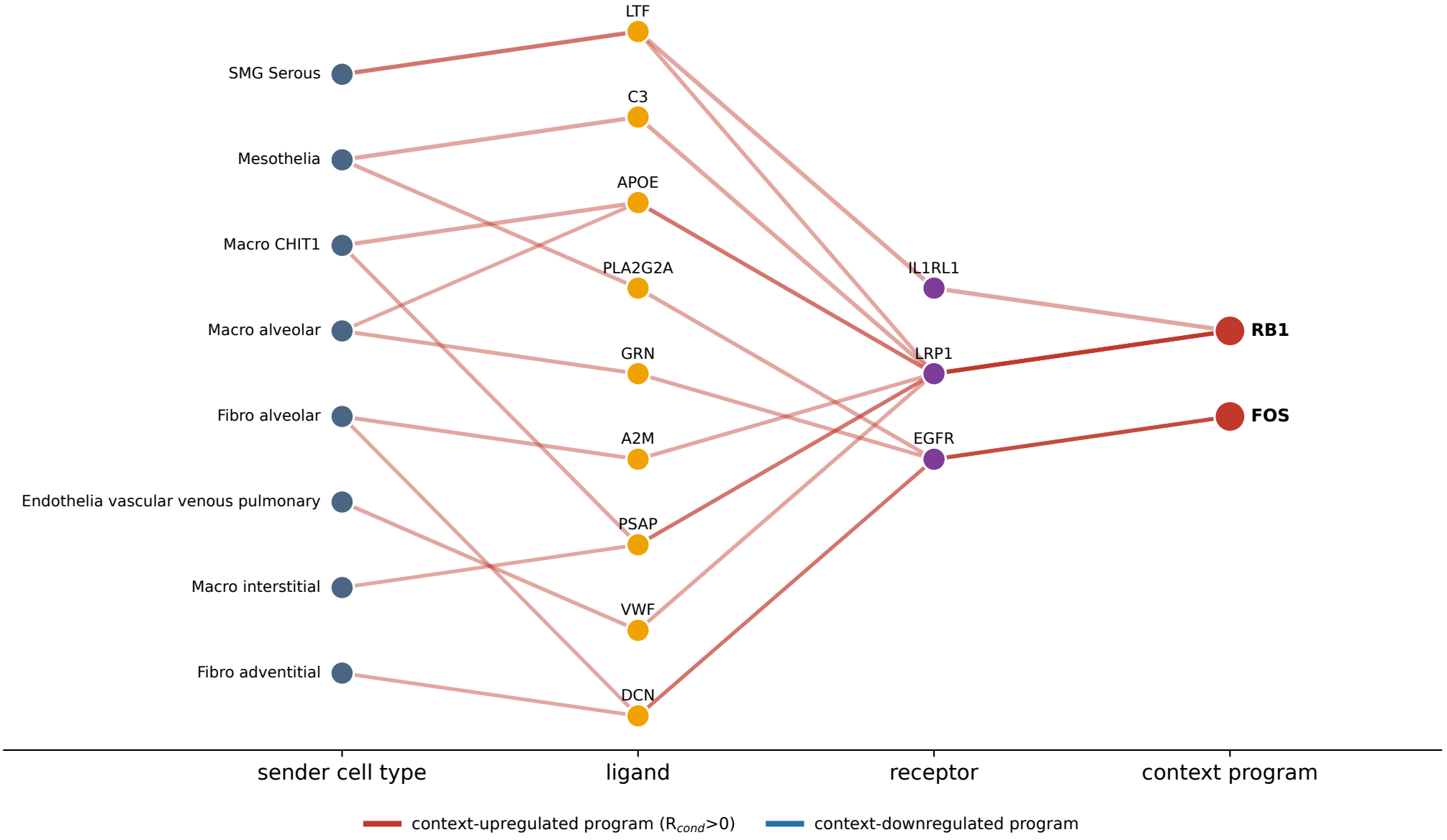

R_{cond}-gated communication chains, AAH → AIS
(top chains from ~1.3M candidates; all terminate on context-driven programs)
SMG Serous
Mesothelia
Macro CHIT1
Macro alveolar
Fibro alveolar
Endothelia vascular venous pulmonary
Macro interstitial
Fibro adventitial
LTF
C3
APOE
PLA2G2A
GRN
A2M
PSAP
VWF
DCN
IL1RL1
LRP1
EGFR
RB1
FOS
sender cell type
ligand
receptor
context program
context-upregulated program (R_{cond}>0)
context-downregulated program